Phase 2:
SRS

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Introduction

1.1 Purpose

To set the system that will be used and standards and expectations to be reached from this project.

1.2 Scope

- Authentication
- Add tasks with its details
- Remove tasks
- Comment on tasks
- Due time for each task if there is one
- Reminders
- Notifications
- Chat
- Profile information
- Add a file
- Grouping to add people to project and view tasks added

1.3 Definitions, Acronyms, and Abbreviations

- Admin: Is the Project creator
- Team Member: Are users added by the Project Admin to do certain tasks told by the Admin
- Task: Will be in a to do list like form to show what needs to be done by the team members

1.4 References

- W3Schools
- Labs covered in section

- Youtube

1.5 Overview

Frameworks are covered in this document, expected outcome, features that will be available, and diagrams to represent what everything does and affects

Overall Description

2.1 Product Perspective

Improves teams coordination and time when working on a project

2.2 Product Functions

- Make a project
- Add tasks
- Invite people to join project
- Only people that join a project can see the tasks
- Remove tasks
- Add comments
- Add due date and time
- Reminder of a task deadline closing in
- Add and remove file
- Notification center
- Chat
- Login to account
- Sign in to make an account and add profile details

2.3 User Classes and Characteristics

1. Project admins
2. Team members of a project under the admin

2.4 Software Stakeholders

1. Project managers
2. Students

2.5 Operating Environment

- Browser like FireFox

- Users must have a Mobile Phone, a Tablet, or a PC
- Users must have an internet connection

2.6 Design and Implementation Constraints

- English
- Flask
- Python
- Html
- Css
- SQL

2.7 User Documentation

Users support which they can talk to using the chat

2.8 Assumptions and Dependencies

Assure all users have a good internet connection, a device that must have a browser installed on it to access our product from the web

Specific Requirements

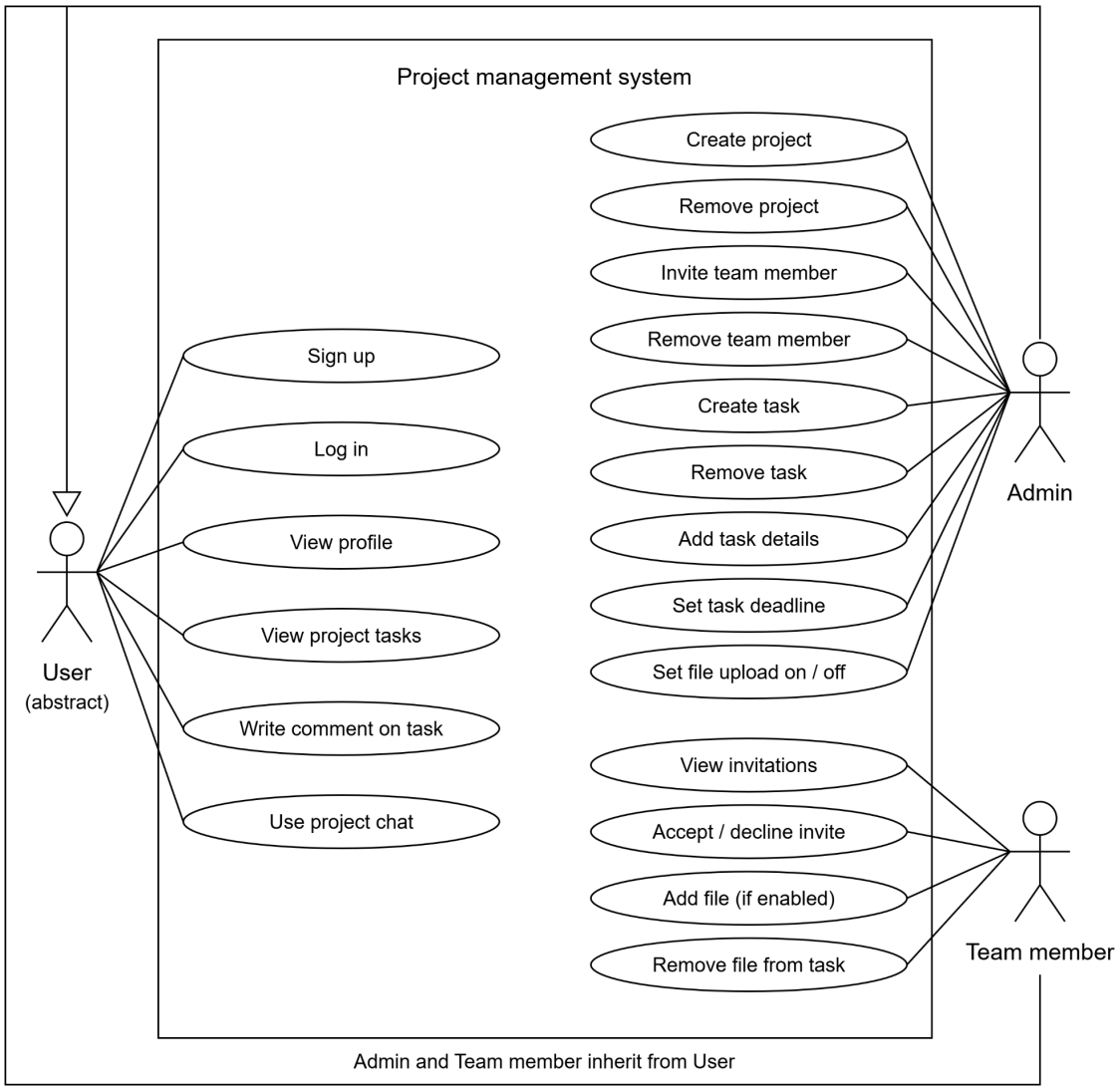
3.1 Functional Requirements

1. Create Project
2. Name Project
3. Add Project description
4. Invite team members to project
5. Make task
6. Add deadline
7. Add task details
8. Add a place to comment on a task
9. Make a check box to mark when task in complete
10. Add a file by team member
11. Remove File by team member
12. Reminder of a tasks deadline closing in
13. Login
14. Sign in and add details like name

15. Go to chat
16. Notification
17. Delete Project
18. Remove a team member
19. Remove file adding in a task if chosen by the project admin
20. Give the ability to add a file by the project admin in a certain task

3.2 Use Case Model

- 3.2.1 Use Case Diagram



- ### 3.2.2 Use Case Descriptions

1. Sign Up

Allows a new user to create an account by providing the required information, such as name, email, and password, to access the Project Management System

2. Log In

Allows registered users to access their accounts by entering their email and password

3. View Profile

Allows users to view their personal profile information, including their name, email, and other profile details

4. View Project Tasks

Allows users who are members of a project to view the tasks belonging to that project and check their details and progress

5. Write Comment on Task

Allows users to write comments on tasks to share feedback, ask questions, or discuss task-related information with other project members

6. Use Project Chat

Allows project users to communicate with other members through a shared project chat

7. Create Project

Allows the admin to create a new project by providing its name and description. The admin becomes the project owner and can manage its tasks and team members

8. Remove Project

Allows the admin to delete a project that is no longer needed

9. Invite Team Member

Allows the admin to invite users to join a project so they can participate in its activities and work on assigned tasks

10. Remove Team Member

Allows the admin to remove a team member from a project when their participation is no longer required

11. Create Task

Allows the admin to create a new task within a project and make it available for the project team to work on

12. Remove Task

Allows the admin to delete an existing task from the project when it is no longer required

13. Add Task Details

Allows the admin to provide additional information about a task, such as its title and description, to clarify what needs to be done

14. Set Task Deadline

Allows the admin to set a due date and time for a task to specify when it should be completed

15. Set File Upload On / Off

Allows the admin to enable or disable the ability of team members to upload files to a specific task

16. View Invitations

Allows team members to view the project invitations they have received from project admins

17. Accept / Decline Invitation

Allows team members to accept or decline invitations to join projects. If an invitation is accepted, the user becomes a member of that project

18. Add File (If Enabled)

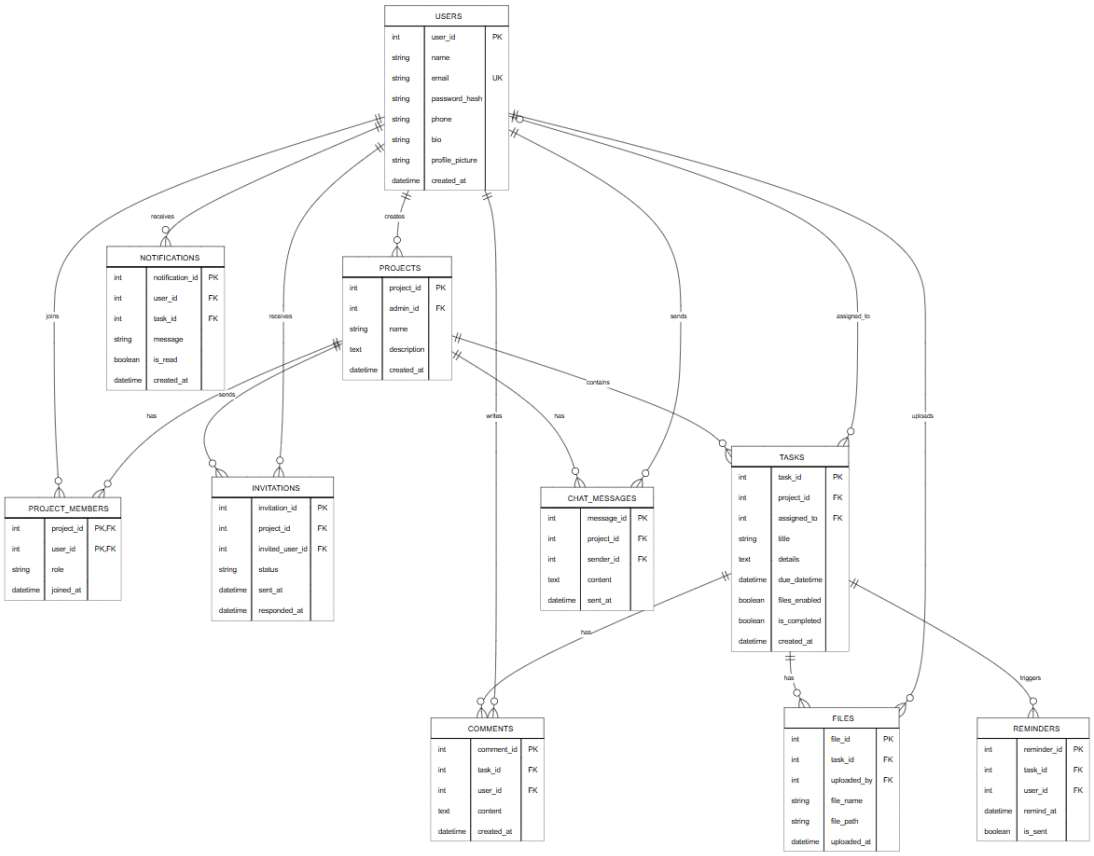
Allows team members to upload files related to a task, provided that the admin has enabled file uploads for that task

19. Remove File from Task

Allows team members to remove files they have added to a task when they are no longer needed

3.3 Domain Model

- ### 3.3.1 Conceptual Class Diagram



- 3.3.2 Class Descriptions

The following classes represent the main entities of the Project Management System. Each class is responsible for storing and managing specific information related to users, projects, tasks, and communication

1. User

Represents a registered user in the system. It stores the user's personal information and account details, such as user ID, name, email, password, phone number, bio, and profile picture

2. Project

Represents a project created and managed by an admin. It stores the project's information, including project ID, admin ID, name, description, and creation date

3. Project Member

Represents the relationship between users and projects. It records which users belong to each project, their roles, and the date they joined the project

4. Invitation

Represents an invitation sent by a project admin to a user to join a project. It stores invitation details, including the project, invited user, invitation status, and relevant dates

5. Task

Represents a task created within a project. It stores task information, including its ID, project ID, assigned user, title, details, due date, completion status, and creation date

6. Comment

Represents a comment written by a user on a specific task. It stores the comment content, the user who created it, the related task, and the creation date

7. Chat Message

Represents a message sent by a user through the project chat. It stores the message content, sender information, related project, and sending date

8. File

Represents a file uploaded to a task by a project member. It stores file information, including file ID, related task, uploader, file name, file path, and upload date

9. Notification

Represents a notification sent to a user to inform them about relevant project activities. It stores the notification message, related user, related task, read status, and creation date

10. Reminder

Represents a reminder associated with a task deadline. It stores the related task, the user receiving the reminder, the reminder date and time, and whether the reminder has been sent

3.4 Non-Functional Requirements

1. Load time and performance of retrieving and uploading data about things like profile information, tasks, projects must be under 2 seconds
2. Server must be active 24 hours a day
3. Ensure role based authorization
4. Ensure tolerance to high traffic rate
5. Ensure that the users experience is smooth and the UI is easy to use and understand

3.5 External Interface Requirements

- 3.5.1 User Interface

Will provide the user with a smooth and easy to use UI experience

- 3.5.2 Hardware Interface

Server must have and SSD to max out its performance when handling data and a Network Interface Card (NIC) or wireless NIC is required to access the internet and reach the web

- 3.5.3 Software Interface

The system must be able to get the date from the internet and use Emails like google email

- 3.5.4 Communication Interface

The system should be able to use an external email source like google email for email based communication

Appendices

4.1 Appendix A: Data Dictionary

USERS Table:

- user_id: A unique number assigned to every user who signs up.
- name: The user's full name, filled out during registration.
- email: The user’s email address, which doubles as their login username. No two accounts can share the same email.
- password_hash: The securely encrypted version of the user's password. The system never stores plain text passwords.
- phone: An optional contact number for the user.
- bio: A short, customizable description where users can introduce themselves.
- profile_picture: The file path pointing to where the user's profile image is saved on the server.
- created_at: The exact timestamp showing when the account was first created.

PROJECTS Table:

- project_id: A unique identifier for each project.
- admin_id: Links the project directly to the user who created it (the Admin).
- name: The title of the project workspace.
- description: A quick summary outlining what the project is about.
- created_at: The exact timestamp showing when the project was created.

PROJECT_MEMBERS Table:

- project_id: The ID of the specific project.
- user_id: The ID of the user assigned to that project.
- role: Defines what the user can do in this workspace (e.g., 'Team Member').
- joined_at: The date and time the user officially became a member of the project.

INVITATIONS Table:

- invitation_id: A unique tracker number for every invite sent.
- project_id: The project the user is being invited to join.

- invited_user_id: The specific user receiving the invite.
- status: The current state of the invite (starts as 'Pending', changes to 'Accepted' or 'Declined').
- sent_at: When the admin sent the invite.
- responded_at: When the user accepted or declined the invitation.

TASKS Table:

- task_id: A unique ID for every single task.
- project_id: The project workspace this task belongs to.
- assigned_to: The team member responsible for finishing the task. Left blank if unassigned.
- title: A short summary of what needs to be done.
- details: In-depth instructions or notes for the task.
- due_datetime: The deadline date and time for the task.
- files_enabled: A simple True/False switch set by the admin to allow or block file uploads for this task.
- is_completed: Toggled to True once a team member checks off the task. Defaults to False.
- created_at: The exact timestamp when the task was generated.

COMMENTS Table:

- comment_id: A unique ID for each individual comment.
- task_id: The specific task where this comment was posted.
- user_id: The member who wrote the comment.
- content: The date and time the comment was posted.

CHAT_MESSAGES Table:

- message_id: A unique ID for every chat message.
- project_id: The project chat room where the message was sent.
- sender_id: The user who sent the message.
- content: The text payload of the chat message.
- sent_at: The timestamp showing exactly when the message was sent.

FILES Table:

- file_id: A unique ID for the uploaded file.
- task_id: The task where this file was attached.
- uploaded_by: The user who uploaded the document.
- file_name: The original name of the file (e.g., srs_draft.pdf).
- file_path: The directory path showing exactly where the physical file is stored safely on the server.
- uploaded_at: The timestamp showing when the upload finished.

NOTIFICATIONS Table:

- notification_id: A unique tracking code for the alert.
- user_id: The specific user who needs to see this alert.
- task_id: An optional link to the task that triggered the notification.
- message: The text shown to the user (e.g., "Ali assigned you a new task").
- is_read: Keeps track of whether the user has clicked on/seen the alert yet. Defaults to False.
- created_at: The time the notification was sent.

REMINDERS Table:

- reminder_id: A unique ID for the reminder hook.
- task_id: The task with the upcoming deadline.
- user_id: The user scheduled to receive the reminder alert.
- remind_at: The exact future date and time this reminder is scheduled to go off.
- is_sen: Tracks whether the system has successfully delivered the reminder to the user.

4.2 Appendix B: Glossary

- **Admin:** The person who creates a project workspace. They have full control over the project, including inviting or removing members, managing tasks, and changing settings.
- **API (Application Programming Interface):** A bridge that allows different software applications to securely talk to each other and share information.
- **CSS (Cascading Style Sheets):** Code used to design the look, layout, colors, and overall visual style of a website.

- **Data Dictionary:** A guide that maps out every table in a database, explaining what kind of information is stored and how the tables link together.
- **Flask:** A lightweight framework for Python used to handle the backend logic, routing, and database communication of a web application.
- **Foreign Key (FK):** A column in a database table that links to a unique key in another table, connecting related data rows together.
- **HTML (HyperText Markup Language):** The structural layout language used to build the basic framework, text fields, and buttons of a web page.
- **Primary Key (PK):** A completely unique identifier (like an ID number) given to a specific row in a database table so it can be easily found and referenced.
- **Project Member:** A registered user who has accepted an invitation to join an admin's project workspace.
- **Python:** The core programming language used to build the backend logic and handle all calculations for this application.
- **Role-Based Authorization:** A security check that ensures users can only perform certain actions (like deleting a project) if they have the right permission role.
- **SQL (Structured Query Language):** The language used to store, update, read, and manage data inside a database.
- **SRS (Software Requirements Specification):** A detailed document that serves as a blueprint for a software project, outlining everything the system must do and how it will operate.
- **Task:** An individual action item in a project that tracks a piece of work to be done. It features details like deadlines, descriptions, and completion checkboxes.